Программный RAID

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Цели и задачи работы

Цель лабораторной работы

Освоить создание, настройку и администрирование RAID-массивов при помощи утилиты mdadm в Linux.

Процесс выполнения лабораторной работы

Проверка дисков

-

```
root@localhost:/home/manieyvazi# whoami
root
root@localhost:/home/manieyvazi# fdisk -l | grep /dev/sd
Disk /dev/sdd: 512.24 MiB, 537122304 bytes, 1049067 sectors
Disk /dev/sdb: 512.26 MiB, 537143296 bytes, 1049108 sectors
Disk /dev/sdc: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk /dev/sda: 40 GiB, 42949672960 bytes, 83886080 sectors
/dev/sda1     2048     4095     2048   1M BIOS boot
/dev/sda2     4096  2101247  2097152   1G Linux extended boot
/dev/sda3  2101248 83884031 81782784  39G Linux LVM
root@localhost:/home/manieyvazi# sfdisk /dev/sdd <<EOF
> ;
> EOF
Checking that no-one is using this disk right now ... OK

Disk /dev/sdd: 512.24 MiB, 537122304 bytes, 1049067 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

>>> Created a new DOS (MBR) disklabel with disk identifier 0xecd219e4.
/dev/sdd1: Created a new partition 1 of type 'Linux' and of size 511.2 MiB.
/dev/sdd2: Done.

New situation:
Disklabel type: dos
Disk identifier: 0xecd219e4

Device     Boot Start     End Sectors   Size Id Type
/dev/sdd1        2048 1049066 1047019 511.2M 83 Linux

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
root@localhost:/home/manieyvazi# sfdisk /dev/sdc <<EOF
> ;
> EOF
Checking that no-one is using this disk right now ... OK

Disk /dev/sdc: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

Рис. 1 — Проверка дисков

Создание разделов

-.

```
root@localhost:/home/manieyvazi# sfdisk --print-id /dev/sdd 1
sfdisk: print-id is deprecated in favour of --part-type
83
root@localhost:/home/manieyvazi# sfdisk --print-id /dev/sdc 1
sfdisk: print-id is deprecated in favour of --part-type
83
root@localhost:/home/manieyvazi# sfdisk --print-id /dev/sdb 1
sfdisk: print-id is deprecated in favour of --part-type
83
root@localhost:/home/manieyvazi# sfdisk -T | grep -i raid
fd  Linux raid autodetect
root@localhost:/home/manieyvazi# sfdisk --change-id /dev/sdd 1 fd
sfdisk: change-id is deprecated in favour of --part-type

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
root@localhost:/home/manieyvazi# sfdisk --change-id /dev/sdc 1 fd
sfdisk: change-id is deprecated in favour of --part-type

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
root@localhost:/home/manieyvazi# sfdisk --change-id /dev/sdb 1 fd
sfdisk: change-id is deprecated in favour of --part-type

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
root@localhost:/home/manieyvazi# sfdisk -l /dev/sdd
Disk /dev/sdd: 512.24 MiB, 537122304 bytes, 1049067 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xecd219e4

Device     Boot Start     End Sectors   Size Id Type
/dev/sdd1        2048 1049066 1047019 511.2M fd Linux raid autodetect
root@localhost:/home/manieyvazi# sfdisk -l /dev/sdc
Disk /dev/sdc: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk model: VBOX HARDDISK
```

Рис. 2 — Типы разделов и изменение типа на RAID

Разделы типа RAID

-

```
root@localhost:/home/manieyvazi# sfdisk -l /dev/sdc
Disk /dev/sdc: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc9cf3cef

Device     Boot Start     End Sectors   Size Id Type
/dev/sdc1        2048 1049353 1047306 511.4M fd Linux raid autodetect
root@localhost:/home/manieyvazi# sfdisk -l /dev/sdb
Disk /dev/sdb: 512.26 MiB, 537143296 bytes, 1049108 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcc213558

Device     Boot Start     End Sectors   Size Id Type
/dev/sdb1        2048 1049107 1047060 511.3M fd Linux raid autodetect
root@localhost:/home/manieyvazi# mdadm --help
mdadm is used for building, managing, and monitoring
Linux md devices (aka RAID arrays)
Usage: mdadm --create device options...
            Create a new array from unused devices.
       mdadm --assemble device options...
            Assemble a previously created array.
       mdadm --build device options...
            Create or assemble an array without metadata.
       mdadm --manage device options...
            make changes to an existing array.
       mdadm --misc options... devices
            report on or modify various md related devices.
       mdadm --grow options device
            resize/reshape an active array
       mdadm --incremental device
            add/remove a device to/from an array as appropriate
       mdadm --monitor options...
            Monitor one or more array for significant changes.
       mdadm device options...
            Shorthand for --manage.
Any parameter that does not start with '-' is treated as a device name
or, for --examine-bitmap, a file name.
The first such name is often the name of an md device.  Subsequent
```

Рис. 3 — Разделы типа fd (Linux raid autodetect)

Создание RAID1

-.

```
 For detailed help on the above major modes use --help after the mode
 e.g.
        mdadm --assemble --help
 For general help on options use
root@localhost:/home/manieyvazi# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2
mdadm: You haven't given enough devices (real or missing) to create this array
root@localhost:/home/manieyvazi# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdd1 /dev/sdc1
mdadm: Note: this array has metadata at the start and
    may not be suitable as a boot device.  If you plan to
    store '/boot' on this device please ensure that
    your boot-loader understands md/v1.x metadata, or use
    --metadata=0.90
mdadm: size set to 522432K
Continue creating array [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@localhost:/home/manieyvazi# cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdc1[1] sdd1[0]
      522432 blocks super 1.2 [2/2] [UU]

unused devices: <none>
root@localhost:/home/manieyvazi# mdadm --query /dev/md0
/dev/md0: 510.19MiB raid1 2 devices, 0 spares. Use mdadm --detail for more detail.
root@localhost:/home/manieyvazi# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:03:40 2025
        Raid Level : raid1
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 2
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:03:46 2025
             State : clean
    Active Devices : 2
   Working Devices : 2
    Failed Devices : 0
     Spare Devices : 0

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : b59edb5c:d9f51d49:c8e8d395:2e58cb44
            Events : 17
```

Рис. 4 — Создание RAID1

Детальная информация массива

-.

```
root@localhost:/home/manieyvazi# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:03:40 2025
        Raid Level : raid1
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 2
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:03:46 2025
             State : clean
    Active Devices : 2
   Working Devices : 2
    Failed Devices : 0
     Spare Devices : 0

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : b59edb5c:d9f51d49:c8e8d395:2e58cb44
            Events : 17
```

Рис. 5 — Информация о RAID1

Файловая система и монтирование

-.

```
root@localhost:/home/manieyvazi# mkfs.ext4 /dev/md0
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 522432 1k blocks and 130560 inodes
Filesystem UUID: c8992040-cdd2-44af-a679-44642a207dc3
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

root@localhost:/home/manieyvazi# mkdir /data
mkdir: cannot create directory '/data': File exists
root@localhost:/home/manieyvazi# cd /data
```

Рис. 6 — Создание ext4 и монтирование

Автомонтирование

-.

```
GNU nano 8.1                                        /etc/fstab

#
# /etc/fstab
# Created by anaconda on Fri Sep 26 13:01:01 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=ecf62878-58a1-470b-a7cb-7da192353e46 /                       xfs     defaults        0 0
UUID=61940581-6552-45db-a269-f893ea8c6dac /boot                   xfs     defaults        0 0
UUID=09b04f01-5e64-4e8c-8f3c-b627431e957b none                    swap    defaults        0 0
/dev/vgdata/lvdata                        /mnt/data               ext4    defaults        1 2
/dev/vggroup/lvgroup                      /mnt/groups             xfs     defaults        1 2
/dev/md0                                  /data/mnt               ext4    defaults        1 2

#UUID=f989c7da-164e-42ad-b4b3-4687d8f43a65 /mnt/data              xfs     defaults        1 2
#UUID=cb353ca7-1f68-4b5a-bae6-64260c049ee3 /mnt/data-ext                  ext4    defaults        1 2
#UUID=63b3ad1f-e32c-47d1-bf0c-08a7ae8b8d38 none                   swap    defaults        1 2
```

Рис. 7 — fstab запись

Сбой одного диска

-.

```
       2       8       17        -      spare   /dev/sdb1
root@localhost:~# mdadm /dev/md0 --fail /dev/sdc1
root@localhost:~# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:13:34 2025
        Raid Level : raid1
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:16:46 2025
             State : clean
    Active Devices : 2
   Working Devices : 2
    Failed Devices : 1
     Spare Devices : 0

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : adad3d5d:b67b1000:4f8ff15d:9a175a09
            Events : 37

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       2       8       17        1      active sync   /dev/sdb1

       1       8       33        -      faulty   /dev/sdc1
root@localhost:~# mount /dev/md0
mount: /dev/md0: can't find in /etc/fstab.
root@localhost:~# mdadm --stop /dev/md0
mdadm: stopped /dev/md0
root@localhost:~# mdadm --zero-superblock /dev/sdd1
root@localhost:~# mdadm --zero-superblock /dev/sdc1
root@localhost:~# mdadm --zero-superblock /dev/sdb1
root@localhost:~# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdd1 /dev/sdc1
mdadm: Note: this array has metadata at the start and
    may not be suitable as a boot device.  If you plan to
    store '/boot' on this device please ensure that
    your boot-loader understands md/v1.x metadata, or use
    --metadata=0.90
mdadm: size set to 522432K
Continue creating array [y/N]? y
```

Рис. 8 — Сбой и удаление устройства

Завершение работы массива

Проверка работы системы.

```
root@localhost:~# mount /dev/md0
mount: /dev/md0: can't find in /etc/fstab.
root@localhost:~# mdadm --stop /dev/md0
mdadm: stopped /dev/md0
root@localhost:~# mdadm --zero-superblock /dev/sdd1
root@localhost:~# mdadm --zero-superblock /dev/sdc1
root@localhost:~# mdadm --zero-superblock /dev/sdb1
root@localhost:~# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdd1 /dev/sdc1
mdadm: Note: this array has metadata at the start and
    may not be suitable as a boot device.  If you plan to
    store '/boot' on this device please ensure that
    your boot-loader understands md/v1.x metadata, or use
    --metadata=0.90
mdadm: size set to 522432K
Continue creating array [y/N]? y
```

Рис. 9 — Остановка и очистка суперблоков

Создание массива и добавление hot spare

-

```
root@localhost:~# whoami
root
root@localhost:~# mdadm --create --verbose /dev/md0 --level=1 --raid-device=2 /dev/sdd1 /dev/sdc1
mdadm: Note: this array has metadata at the start and
    may not be suitable as a boot device.  If you plan to
    store '/boot' on this device please ensure that
    your boot-loader understands md/v1.x metadata, or use
    --metadata=0.90
mdadm: size set to 522432K
Continue creating array [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@localhost:~# mdadm --add /dev/md0 /dev/sdb1
mdadm: added /dev/sdb1
root@localhost:~# mount /dev/md0
mount: /dev/md0: can't find in /etc/fstab.
root@localhost:~# cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdb1[2](S) sdc1[1] sdd1[0]
      522432 blocks super 1.2 [2/2] [UU]

unused devices: <none>
root@localhost:~# mdadm --query /dev/md0
/dev/md0: 510.19MiB raid1 2 devices, 1 spare. Use mdadm --detail for more detail.
root@localhost:~# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:13:34 2025
        Raid Level : raid1
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:14:07 2025
             State : clean
    Active Devices : 2
   Working Devices : 3
    Failed Devices : 0
     Spare Devices : 1

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
```

Рис. 10 — Hot spare добавлен

Состояние массива с резервом

•

```
unused devices: <none>
root@localhost:~# mdadm --query /dev/md0
/dev/md0: 510.19MiB raid1 2 devices, 1 spare. Use mdadm --detail for more detail.
root@localhost:~# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:13:34 2025
        Raid Level : raid1
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:14:07 2025
             State : clean
    Active Devices : 2
   Working Devices : 3
    Failed Devices : 0
     Spare Devices : 1

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
```

Рис. 11 — Состояние RAID с hotspare

Сбой устройства

•

```
root@localhost:~# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:19:24 2025
        Raid Level : raid5
        Array Size : 1044864 (1020.38 MiB 1069.94 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 3
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:23:41 2025
             State : clean
    Active Devices : 3
   Working Devices : 3
    Failed Devices : 0
     Spare Devices : 0

            Layout : left-symmetric
        Chunk Size : 64K

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : 7a1b7858:c62664cb:7f8c38dc:309f0c1d
            Events : 37

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       33        1      active sync   /dev/sdc1
       2       8       17        2      active sync   /dev/sdb1
root@localhost:~# umount --stop /dev/md0
umount: unrecognized option '--stop'
Try 'umount --help' for more information.
root@localhost:~# umount /dev/md0
umount: /dev/md0: not mounted.
root@localhost:~# mdadm --stop /dev/md0
mdadm: stopped /dev/md0
root@localhost:~# mdadm --zero-superblock /dev/sdd1
root@localhost:~# mdadm --zero-superblock /dev/sdc1
root@localhost:~# mdadm --zero-superblock /dev/sdb1
root@localhost:~# /dev/md0 /data1 ext4 defaults 1 2
bash: /dev/md0: No such file or directory
root@localhost:~#
```

Рис. 12 — Активация резервного диска

Исходный RAID1 перед конвертацией

•

```
root@localhost:~# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdd1 /dev/sdc1
mdadm: Note: this array has metadata at the start and
    may not be suitable as a boot device.  If you plan to
    store '/boot' on this device please ensure that
    your boot-loader understands md/v1.x metadata, or use
    --metadata=0.90
mdadm: size set to 522432K
Continue creating array [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@localhost:~# mdadm --add /dev/md0 /dev/sdb1
mdadm: added /dev/sdb1
root@localhost:~# mount /dev/md0
mount: /dev/md0: can't find in /etc/fstab.
root@localhost:~# cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdb1[2](S) sdc1[1] sdd1[0]
      522432 blocks super 1.2 [2/2] [UU]

unused devices: <none>
root@localhost:~# mdadm --query /dev/md0
/dev/md0: 510.19MiB raid1 2 devices, 1 spare. Use mdadm --detail for more detail.
root@localhost:~# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:19:24 2025
        Raid Level : raid1
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:20:08 2025
             State : clean
    Active Devices : 2
   Working Devices : 3
    Failed Devices : 0
     Spare Devices : 1

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : 7a1b7858:c62664cb:7f8c38dc:309f0c1d
            Events : 18
```

Рис. 13 — Перед преобразованием

Состояние массива

•

```
    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       33        1      active sync   /dev/sdc1

       2       8       17        -      spare   /dev/sdb1
root@localhost:~# mdadm --grow /dev/md0 --level=5
mdadm: level of /dev/md0 changed to raid5
root@localhost:~# mdadm --detail /dev/md9
mdadm: cannot open /dev/md9: No such file or directory
root@localhost:~# mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Dec 17 07:19:24 2025
        Raid Level : raid5
        Array Size : 522432 (510.19 MiB 534.97 MB)
     Used Dev Size : 522432 (510.19 MiB 534.97 MB)
      Raid Devices : 2
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Dec 17 07:22:07 2025
             State : clean
    Active Devices : 2
   Working Devices : 3
    Failed Devices : 0
     Spare Devices : 1

            Layout : left-symmetric
        Chunk Size : 64K

Consistency Policy : resync

              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : 7a1b7858:c62664cb:7f8c38dc:309f0c1d
            Events : 19

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       33        1      active sync   /dev/sdc1

       2       8       17        -      spare   /dev/sdb1
root@localhost:~# mdadm --grow /dev/md0 --raid-devices 3
```

Рис. 14 — Состояние RAID перед изменением

Изменение уровня массива

-

```
              Name : localhost.localdomain:0  (local to host localhost.localdomain)
              UUID : b59edb5c:d9f51d49:c8e8d395:2e58cb44
            Events : 17

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       33        1      active sync   /dev/sdc1
root@localhost:/home/manieyvazi# mkfs.ext4 /dev/md0
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 522432 1k blocks and 130560 inodes
Filesystem UUID: c8992040-cdd2-44af-a679-44642a207dc3
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

root@localhost:/home/manieyvazi# mkdir /data
mkdir: cannot create directory '/data': File exists
root@localhost:/home/manieyvazi# cd /data
root@localhost:/data# ls
main  third
root@localhost:/data# cd
root@localhost:~# mkdir /data1
root@localhost:~# mount /dev/md0 /data1
root@localhost:~# /dev/md0 /data1 ext4 defaults 1 2
bash: /dev/md0: Permission denied
root@localhost:~# whoami
root
root@localhost:~# /dev/md0 /data1 ext4 defaults 1 2
bash: /dev/md0: Permission denied
root@localhost:~# mdadm /dev/md0 --fail /dev/sdc1
root@localhost:~# mdadm /dev/md0 --remove /dev/sdc1
mdadm: hot removed /dev/sdc1 from /dev/md0
root@localhost:~# mdadm /dev/md0 --add /dev/sdb1
mdadm: added /dev/sdb1
root@localhost:~# umount /dev/md0
root@localhost:~# mdadm --stop /dev/md0
mdadm: stopped /dev/md0
root@localhost:~# mdadm --zero-superblock /dev/sdd1
root@localhost:~# mdadm --zero-superblock /dev/sdb1
root@localhost:~# mdadm --zero-superblock /dev/sdc1
root@localhost:~# whoami
root
```

Рис. 15 — Преобразование в RAID5

Финальная конфигурация RAID5

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Рис. 16 — Итоговый RAID5

Выводы по проделанной работе

Вывод

В ходе работы были изучены принципы создания и администрирования RAID-массивов в Linux. Созданы и протестированы конфигурации RAID 1, RAID 1 с hot spare, а также выполнено преобразование массива RAID 1 в RAID 5 и расширение его состава.